

Lab Tasks

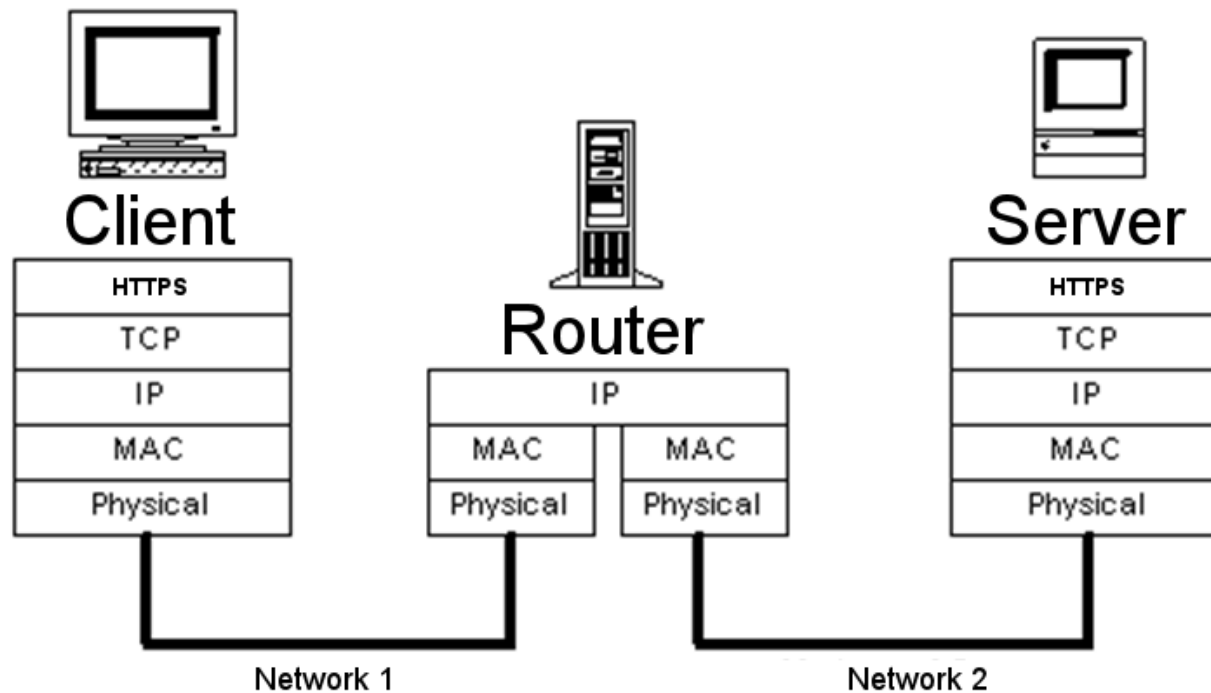
MI102 - Network Systems and Security

Fachhochschule Kiel

Florian Caspar

Task 2 (1)

- Network architecture overview:



Task 2 (2)

- All tasks need to be done twice, once for Windows environment and once for Linux environment.

Task 2 (3)

- Configure all VMs to operate only in the VM environment. The three devices Client, Router and Server shall not be able to reach other devices outside Network 1 and Network 2.

Task 2 (4)

- Install the following software components:
 - Router:
 - Firewall-Application
 - DNS-Server
 - Server:
 - Webserver

Task 2 (5)

- Perform the following configuration tasks:
 - Client
 - shall use Router as DNS-Server

Task 2 (6)

- Perform the following configuration tasks:
 - Router
 - Firewall
 - shall do port forwarding for protocols HTTP and HTTPS between Client and Server
 - shall open ports for accessing local DNS-Server
 - DNS-Server
 - create all necessary entries for Webserver

Task 2 (7)

- Perform the following configuration tasks:
 - Server
 - Webserver
 - set up a simple example HTML document
 - install an HTTPS certificate
 - server shall be reachable via HTTP and HTTPS
 - all HTTP-requests shall be rewritten to HTTPS requests
 - HSTS shall be active
 - Local Firewall
 - shall open ports for accessing local Webserver

Task 2 (8)

- Checklist for completion of lab task:
 - each for Windows and Linux
 - Client can request example HTML document from Webserver using server host name and HTTPS
 - Client can request any document from Webserver using HTTP and is forwarded to same document using HTTPS
 - Client receives always HSTS header from Webserver
 - Port Scan of Router from the Client shows proper configuration of firewall.
 - Port Scan of Webserver from the Router shows proper configuration of local firewall.

©2021 Florian Caspar